Cover letter — EPSL submission

Dear Editors,

We submit the enclosed Letter, 'A network framework for comparing circulation architectures across planetary bodies'. Planetary activity is normally described process by process or collapsed informally onto an active–inactive continuum; whether that continuum is adequate has never been tested on a common measurement basis.

We built a frozen seven-reservoir, seven-process ontology and applied it as three separated data objects — current activity, persistence through geological history, and observational constraint — to Earth, Venus, Mars, the Moon, Mercury, Io, Europa, and Titan. The principal empirical results: comparative state retains a dominant first-order activity gradient, with ~44% of ordination variance on other axes; subsystem shutdown is asynchronous where it occurs (Mars lost its dynamo ~4 Ga before its latest volcanism while its surface–atmosphere circulation persists); and supported-process counts are uncorrelated with age or mass but organized by energy-source architecture.

The result matters beyond the eight bodies: it offers comparative planetology a common, falsifiable state-space representation at a moment when exoplanet demography makes such comparisons pressing. The concise, quantitative, cross-disciplinary format is designed for EPSL's readership. All data and code are public at https://github.com/bougtoir/planetary_circulation.

The manuscript is original, not under consideration elsewhere, and all authors [TO COMPLETE] approve submission. We have no competing interests to declare [VERIFY].

Sincerely, [CORRESPONDING AUTHOR — TO COMPLETE]